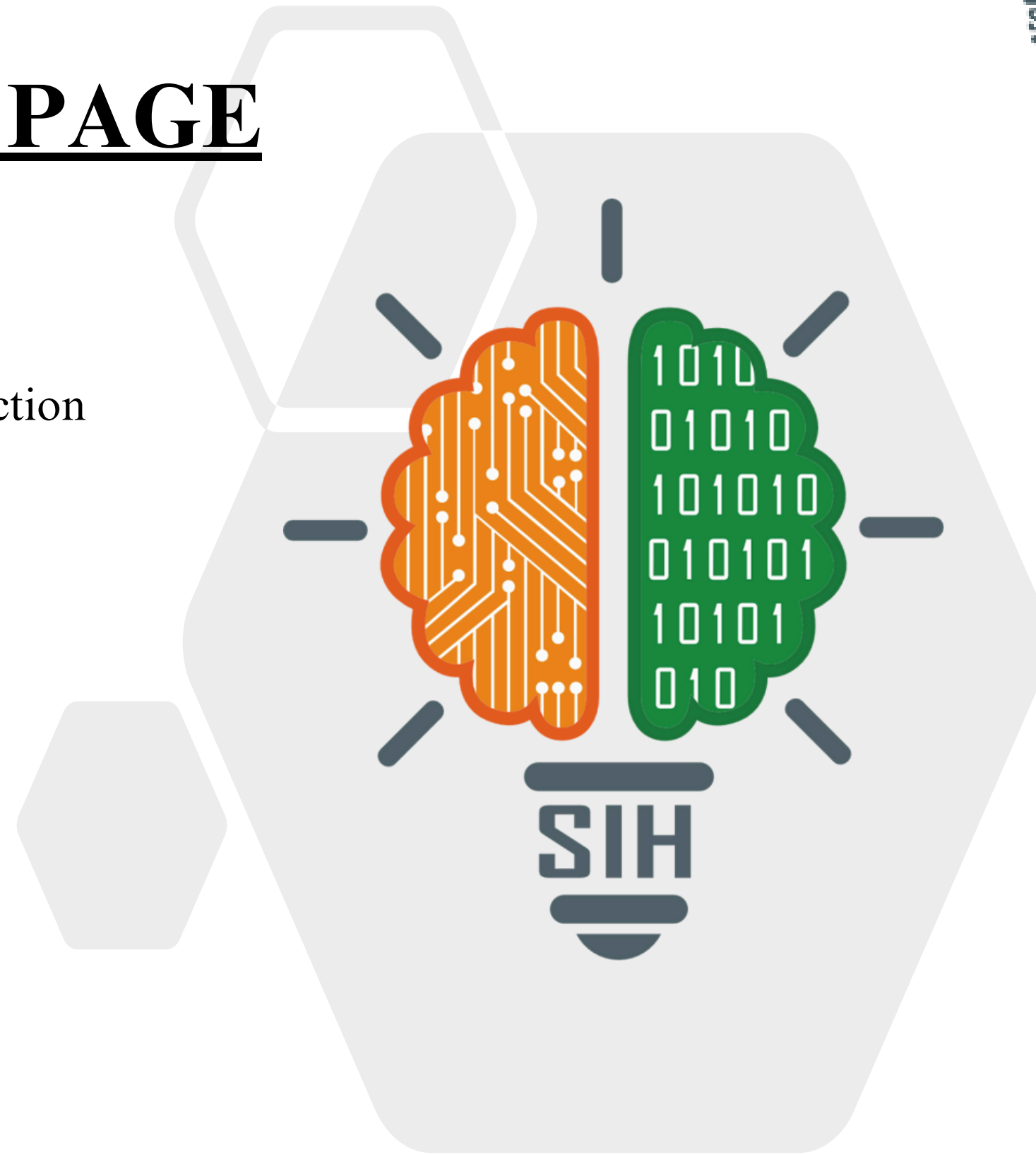


TITLE PAGE

- **Problem Statement ID** –25036
- **Problem Statement Title**-Development of Sensor for Detection of Microplastics
- **Theme**-Miscellaneous
- **PS Category**- Hardware
- **Team ID**-83034
- **Team Name (Registered on portal)**-MicroTrackerss



Microplastic Detection System

Key Features

Technologies and Implementation



Real-time Output

Outputs real-time concentration & type of microplastics.

AI Classification

AI models (LDA, SVM) classify particles as plastic / non-plastic.

Signal Conversion

Photodetectors + DAQ convert signals for analysis.

Parameter Extraction

Polarization analyzer extracts unique parameters (q, u, v) as "fingerprints".

Light Scattering

Polarized light scatters differently based on particle size, shape & material.

Microplastic Detection

Sensor detects microplastics using polarized light scattering + AI.

Modular IoT Design

Modular, IoT-ready design → scalable for remote monitoring.

Real-Time Operation

Real-time, on-site operation with minimal prep.

Polarization Parameters

Uses polarization parameters for higher discrimination accuracy.



Microplastic Differentiation

Differentiates microplastic types & degradation states.



AI-Driven Classification

AI-driven classification for rapid & reliable results.

Programming & Frameworks

Python, Scikit-learn, NumPy, Pandas for AI/data processing. Embedded C/C++, Arduino/Raspberry Pi OS for hardware control.

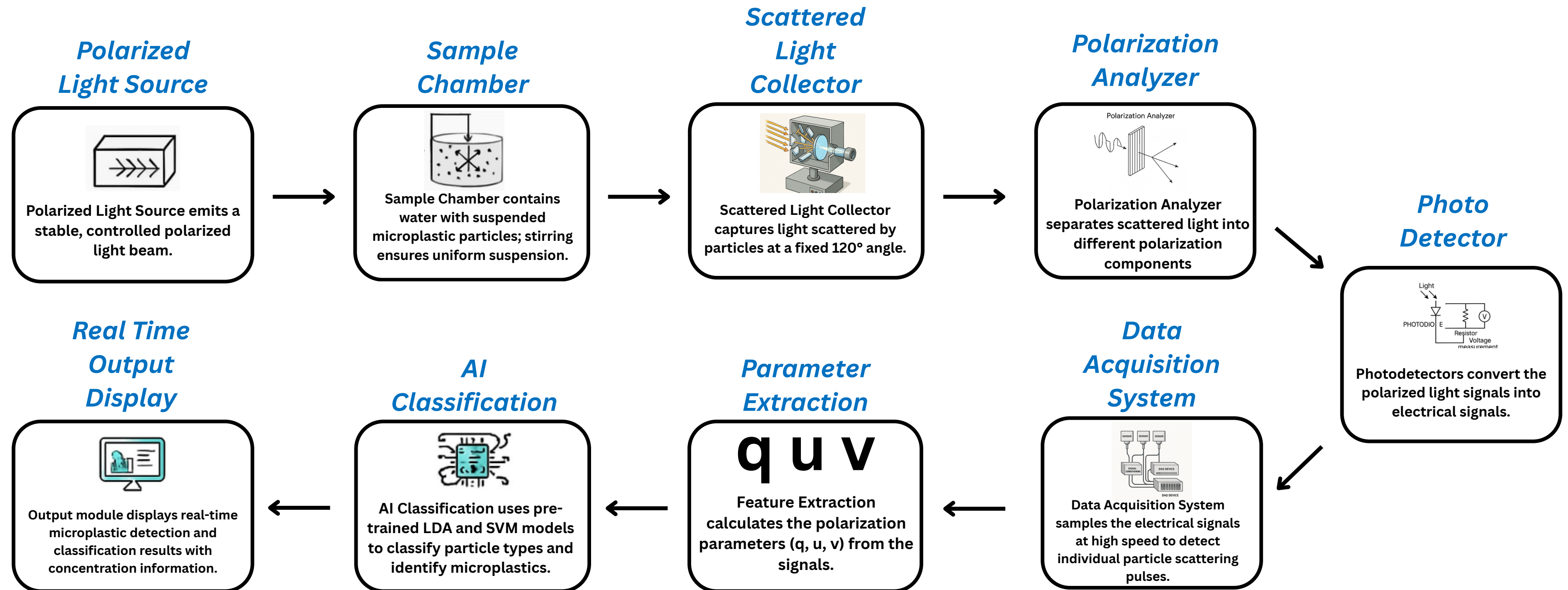


Hardware

Polarized Laser/LED, Optical Components, Sample Chamber with Stirring. Photodetectors, DAQ System, Microcontroller, Communication Module.

TECHNICAL APPROACH

Micro
Trackerss



Feasibility (Technical & Practical)



Polarized light scattering + AI

LDA/SVM capable of microplastic detection.



Particle Detection

Detects particles $\sim 0.5 - 500 \mu\text{m}$, lab accuracy 80–95%.



Components Available

laser, optics, photodiodes, microcontroller.



Prototype

Possible, portable, low-power.

Viability



Long-Term Use & Impact

Useful for lab, field, and water quality monitoring applications.



Real-time Detection

Supports real-time detection & classification (plastic vs non-plastic).



Reliability Needs

Needs calibration, anti-fouling design, & diverse training data for reliability.

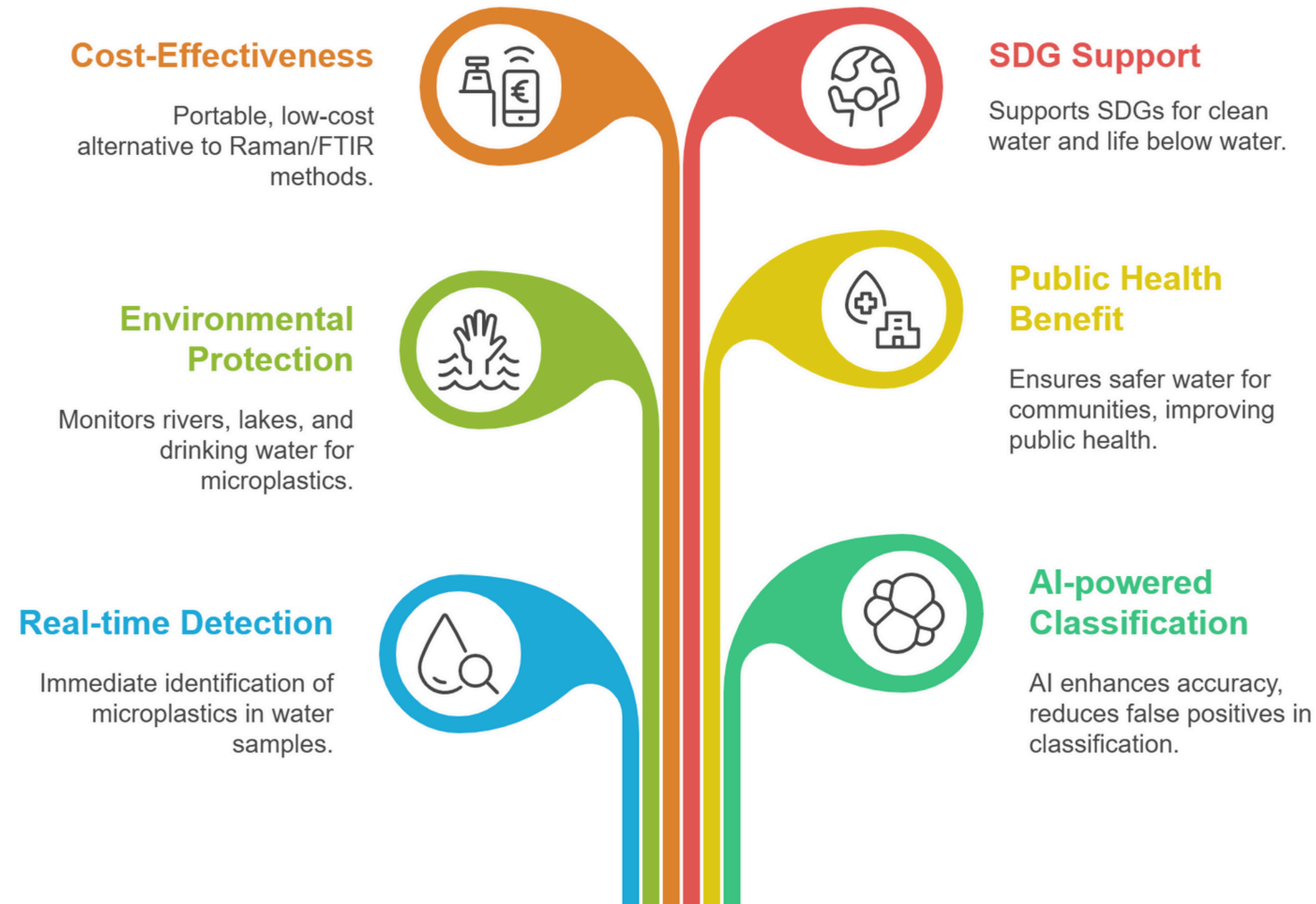


Commercialization

Commercialization possible after improving ruggedness & lowering cost.

IMPACT AND BENEFITS

Unveiling the Multifaceted Benefits of Microplastic Detection



- <https://www.frontiersin.org/journals/marine-science/articles/10.3389/fmars.2021.739683/full>
- <https://pubs.rsc.org/en/content/articlehtml/2022/ay/d2ay01215d>
- <https://pmc.ncbi.nlm.nih.gov/articles/PMC7956778/>
- <https://www.sciencedirect.com/science/article/pii/S0045653519310203>
- https://en.wikipedia.org/wiki/Linear_discriminant_analysis
- https://en.wikipedia.org/wiki/Support_vector_machine